scientific reports

OPEN

Classification of melanocytic lesions using direct illumination multispectral imaging

Elisabeth Victoria Goessinger[1,9], Paul-Gerald Dittrich[2,3,4,9]✉, Philipp Nöcker[2], Gunther Notni[4], Sebastian Weber[5], Sara Cerminara[1], Beda Mühleisen[1], Alexander A. Navarini[1] & Lara Valeska Maul[1,6,7,8]✉

With rising melanoma incidence and mortality, early detection and surgical removal of primary lesions is essential. Multispectral imaging is a new, non-invasive technique that can facilitate skin cancer detection by measuring the reflectance spectra of biological tissues. Currently, incident illumination allows little light to be reflected from deeper skin layers due to high surface reflectance. A pilot study was conducted at the University Hospital Basel to evaluate, whether multispectral imaging with direct light coupling could extract more information from deeper skin layers for more accurate dignity classification of melanocytic lesions. 27 suspicious pigmented lesions from 23 patients were included (6 melanomas, 6 dysplastic nevi, 12 melanocytic nevi, 3 other). Lesions were imaged before excision using a prototype snapshot mosaic multispectral camera with incident and direct illumination with subsequent dignity classification by a pre-trained multispectral image analysis model. Using incident light, a sensitivity of 83.3% and a specificity of 58.8% were achieved compared to dignity as determined by histopathological examination. Direct light coupling resulted in a superior sensitivity of 100% and specificity of 82.4%. Convolutional neural network classification of corresponding red, green, and blue lesion images resulted in 16.7% lower sensitivity (83.3%, 5/6 malignant lesions detected) and 20.9% lower specificity (61.5%) compared to direct light coupling with multispectral image classification. Our results show that incorporating direct light multispectral imaging into the melanoma detection process could potentially increase the accuracy of dignity classification. This newly evaluated illumination method could improve multispectral applications in skin cancer detection. Further larger studies are needed to validate the camera prototype.

Keywords Multispectral imaging, Filter-on-chip, Snapshot-mosaic, Skin cancer, Melanoma, Pilot study

Global incidence of melanoma is rapidly increasing in fair-skinned populations[1]. The best prognosis is based on early detection and surgical removal of primary lesions[2]. The current gold standard for screening is a dermatological full-body examination using dermoscopy, but new technological approaches are needed to optimize detection rates. While failure to diagnose a melanoma early can be detrimental, unnecessary excision of benign lesions increases morbidity and healthcare costs. Multiple non-invasive optical technologies have been investigated to improve skin cancer diagnosis, but have either reached their limits or are not used routinely[3].

Multispectral imaging devices have promising potential to facilitate the detection and dignity assessment of melanocytic skin lesions by providing absorption data hidden to the human eye[4–6]. This approach uses electromagnetic radiation in the visible and the near-infrared spectrum (400–2500 nm) to measure the reflectance spectra of biological tissues and determine lesion dignity based on spectral characteristics[7]. Characteristic absorption bands of lesional skin chromophores such as melanin, de- and oxygenated hemoglobin, or water

[1]Department of Dermatology, University Hospital of Basel, Basel, Switzerland. [2]Imaging and Sensing Department, Fraunhofer Institute for Applied Optics and Precision Engineering IOF, Albert-Einstein-Strasse 7, 07745 Jena, Germany. [3]SpectroNet c/o Technologie- und Innovationspark Jena GmbH, Jena, Germany. [4]Group for Quality Assurance and Industrial Image Processing, Department of Mechanical Engineering, Technische Universität Ilmenau, Ilmenau, Germany. [5]FotoFinder Systems GmbH, Bad Birnbach, Germany. [6]Department of Dermatology, University Hospital Zurich, Zurich, Switzerland. [7]Faculty of Medicine, University of Zurich, Zurich, Switzerland. [8]Department of Dermatology, Felix Platter Hospital, University Hospital Basel, Burgfelderstrasse 101, 4055 Basel, Switzerland. [9]These authors contributed equally: Elisabeth Victoria Goessinger and Paul-Gerald Dittrich. ✉email: paul-gerald.dittrich@iof.fraunhofer.de; lara.maul-duwendag@usz.ch

can be correlated to multispectral imaging analysis results of histologically confirmed skin lesions to provide a differential diagnosis[4].

The underlying principles of this technology for in vivo medical applications have been extensively reviewed[8], but only few have focused on hyperspectral imaging use for melanoma detection[5]. Since melanomas are better vascularized compared to healthy surrounding skin[9], multispectral imaging can noninvasively identify malignancy by estimating hemoglobin and melanin content[10]. The system setup, including the illumination mode, is critical to achieve a wide imaging area to classify the dignity of skin lesions based on their spectral signature[11]. A standard illumination configuration in multispectral imaging systems includes a reliable light source, proper illumination geometry, uniform distribution of light, and environmental regulation to ensure the acquisition of high-quality spectral data. Broadband sources such as halogen lamps or adjustable sources like LEDs are frequently employed to encompass the desired spectral range. Maintaining consistent illumination is essential to mitigate fluctuations in the recorded data. Previously described approaches utilize incident illumination with the illumination plane at the level of the last optical component (e.g., the lens)[12,13]. This results in significant light reflection from the skin surface, comparable to an ultrasound examination without contact gel. By aligning the light source with the skin, direct coupling reduces distracting reflections on the surface, allowing more light to penetrate into deeper layers and extract more information, depending on the wavelength range (Fig. 1)[14].

To our knowledge, data addressing or optimizing illumination mode in multispectral imaging for skin cancer detection is lacking. Using a snapshot-mosaic camera prototype, our study aims to evaluate whether changing the illumination mode from incident to direct light can add significant diagnostic value in the context of melanoma detection.

Materials and methods

Study design and participants

This pilot study was performed at the University Hospital Basel between February and August 2022. Twenty-four patients were recruited in the dermatological outpatient clinic after a routine skin cancer screening led to the decision to excise a pigmented skin lesion to exclude melanoma. Thirty lesions were imaged before excision in incident and direct illumination modalities with varying exposure times using a snapshot-mosaic camera prototype. Only patients aged ≥ 18 years with Fitzpatrick skin types I–III and lesions < 20 mm and > 5 mm in diameter were included. Lesions located on infected and/or injured skin, non-plane surfaces, or mucous membranes were excluded. Additionally, excessive consumption of medications affecting vascularization, heavy physical exercise, substance abuse (including alcohol), cardiovascular disease affecting peripheral vascularization or pregnancy were exclusion criteria.

The study protocol followed the Declaration of Helsinki and was approved by the local Ethics committee (2021-02292). All patients provided written informed consent prior to participation.

Procedure

Before imaging, questionnaires were completed by both physicians and patients. The physician's questionnaire included reporting patient Fitzpatrick skin type, current blood oxygen saturation (spO2), risk of developing skin cancer with further specification (family history, personal history of melanoma, presence of > 5 atypical nevi or > 100 nevi) and listing concomitant dermatological conditions. Patients reported their age, sex, smoking status, pregnancy status, current intake of hormonal or vasoactive medications, and personal history of melanoma.

Lesions were imaged using a prototype of a spatially and spectrally resolving snapshot-mosaic camera, specifically designed for this pilot study. The technical setup and details of the camera are provided in the following section. Both the patient and investigator wore infrared protection goggles throughout the procedure. Initially,

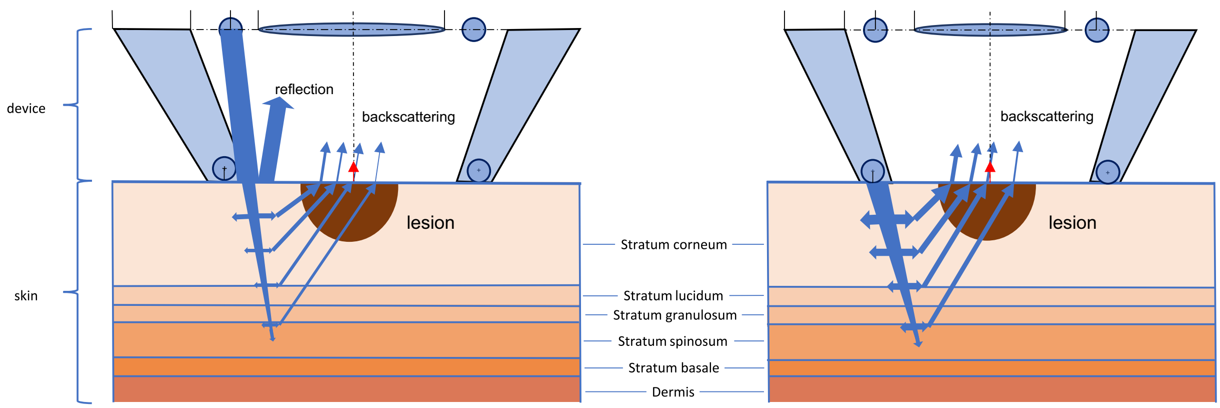

Figure 1. Comparison of illumination methods (left—incident; right—direct)[15]. In conventional multispectral imaging, the light source is positioned at the level of the final optical component (e.g. the lens), maintaining a distance from the skin surface (left image). To enhance the penetration of light into deeper skin structures and reduce backscattering, the illumination source can be aligned parallel to the skin surface (right image). This method is anticipated to produce a more informative reflection profile, yielding valuable insights into the subsurface characteristics of the skin.

any excessive body hair present around lesional skin was shaved to minimize light interference and enhance sensor detection. The location of the lesion of interest was documented using a schematic patient phantom model in the FotoFinder Universe (Version 3.3.1.0., FotoFinder Systems GmbH, Bad Birnbach, Germany, https://www.fotofinder.de/).

Multispectral image acquisition and system control of the prototype were performed using a custom software developed by the Technical University of Ilmenau. The camera was placed directly on the skin. For each trigger, the camera (Fig. 2A) produces a red, green and blue (RGB) image (Fig. 2B) and a 9-wavelength multispectral image (Fig. 2C). For multispectral imaging, manual switching between incident and direct light coupling was possible. In incident illumination mode, a minimum of one image was acquired with an exposure time of 100 ms, with additional images taken at varying exposure times at the investigator's discretion. For direct illumination images, the exposure time was adjusted according to specific skin properties, particularly melanin content, to compensate for the influence of skin characteristics on the image result. Direct light coupling images were captured using nine pre-defined light exposure durations ranging from 100 to 300 ms.

After imaging, active study participation ended, and lesion excision and histopathological examination were performed as part of the clinical routine (Fig. 2D). The final histopathological diagnosis (ground truth) and image data were compared for direct apparent correlations (Fig. 2E). Additionally, downstream hyperspectral image processing was performed for further associative inference, as described in the following sections. Finally, dignity classification based on spectral characteristics was compared to classification performance of RGB images using a commercially-available convolutional neural network (CNN) (MoleAnalyzer-Pro, Version 6.0.5, FotoFinder ATBM® Systems GmbH, Bad Birnbach, Germany, https://www.fotofinder.de/).

Spatially and spectrally resolving snapshot-mosaic camera

The spatially and spectrally resolving snapshot-mosaic camera prototype was developed specifically for this study (Fig. 2A). The camera uses Fabry–Pérot interference filters applied directly in the CMOS-sensor manufacturing process, and a Ruby EV76C661 CMOS sensor. The illumination module for the core function of the direct light coupling method was developed by Tailorlux GmbH (Münster, Germany). The camera has 9 spectral channels, covering a usable wavelength range from 549 to 833 nm. Due to the snapshot-mosaic arrangement of the spectral filters directly on the camera sensor surface, all 9 spectral channels are recorded simultaneously with one snapshot. The pixels are arranged periodically as a 3×3 matrix over the entire CMOS-sensor surface. Each pixel is covered with a filter and is thus assigned to a specific wavelength. All pixels of one wavelength are then combined to form a spectral subimage.

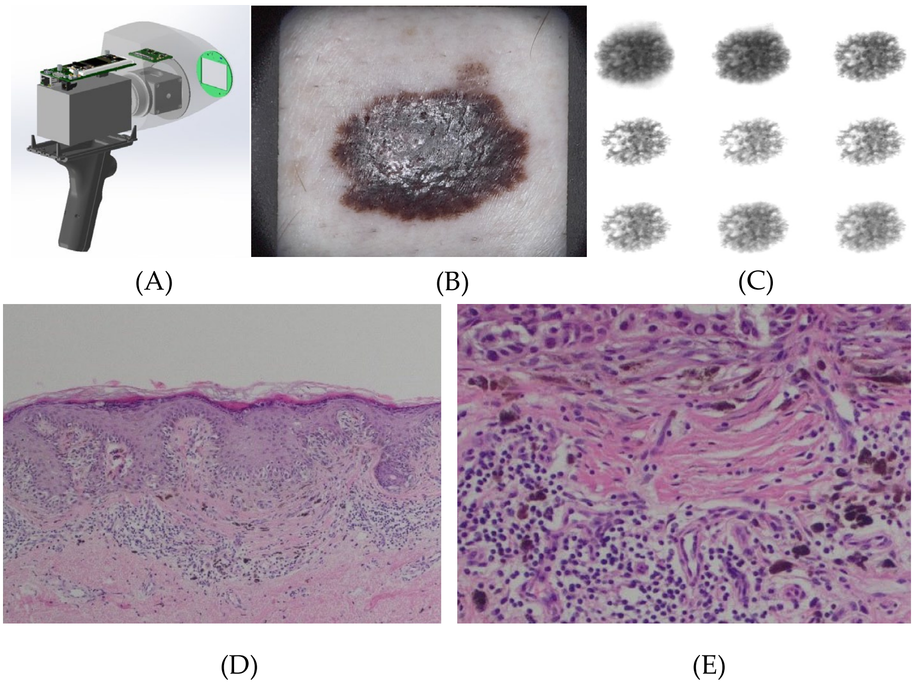

Figure 2. (**A**) Snapshot-mosaic camera 3D model; (**B**) RGB image; (**C**) Sub-images of multispectral image with direct light coupling; (**D**) Superficial spreading melanoma, pT1a, Breslow depth 0.5 mm. HE, $\times 100$ magnification. (**E**) Area with maximum regression (HE, $\times 400$ magnification) corresponding to the shiny white area on the left as seen on RGB and multispectral images.

Preliminary camera calibration and raw data correction

Before clinical study initiation, an extended camera characterization was performed[16,17]. The provided .xml calibration file of the snapshot-mosaic camera was updated with the measured spectral sensitivity curves (Fig. 3A).

After calculation of the reflectance image, demosaicing was performed by the software environment shown in Fig. 3B, resulting in 9 sub-images with an individual size of 340 × 425 pixels. The resulting spatial resolution is therefore limited. Acquisition with a snapshot-mosaic camera has the advantage that all spectral channels are recorded simultaneously. This means that the object only must be in the field of view (FOV) of the camera at one timepoint. A possible rotation of the objects while passing the FOV does not affect the spectral evaluation compared to spectral scanning methods. In our setup, the spatial orientation of the object is not changed while passing the FOV.

Data processing and model building

Multivariate data analysis and processing with spectral model building was performed using fluxTrainer (Version 4.10.6.6, LuxFlux GmbH, Reutlingen, Germany, https://www.luxflux.de/). The acquired multispectral images were used as input together with measured intensities. First, the loaded data was preprocessed using Akima interpolation and a Savitzky-Golay smoothing[19,20]. For spectral processing, spectral endmembers (or signatures) were extracted in a supervised manner. Endmembers were determined by calculating the average spectral signatures of pixel groups formed through clustering based on similar spectral properties. Three reliably classified malignant and benign lesions from pathohistological investigations were used for this purpose. The pixels used for training were excluded from the testing data. To better differentiate the skin components, the spectral false color was calculated using endmembers and their abundances. A distance classifier was used for classification, which calculates the Euclidean distance between a pixel and the group average. For a reliable result, the classified pixels were grouped as objects according to selected parameters. These detected objects are labelled as malignant or benign.

For the illumination mode comparison, two models were built using the same procedure described above. One model used direct- and the other incident light mode data. Each model was fed with small, labeled regions (“malignant”, “benign”, and “healthy skin”) for training. The training regions contained the same coordinates in both datasets to be able to ensure a reliable comparison. The pixel-wise classification allowed us to use the lesions also for testing the model, since the pixels used for training were excluded from the testing data.

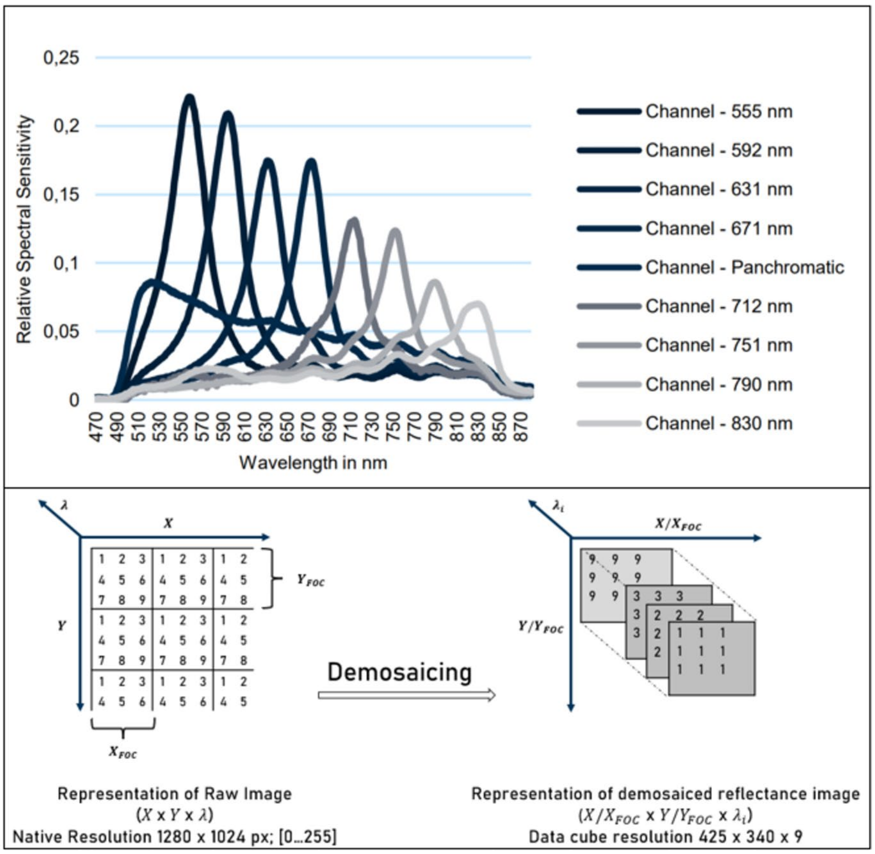

Figure 3. Preliminary camera characterization and calibration. (**A**) Characterized spectral sensitivity curves[18]. (**B**) Resolution and value range of snapshot-mosaic camera images.

Multispectral analysis
The classification process is visualized in Fig. 4A–E. Each pixel is classified individually by the applied model, producing a color map (Fig. 4D). To detect malignant and benign areas, pixel regions with the same classification are grouped together as objects. This facilitates the determination of object parameters and statistical analysis.

Statistical analysis
The exploratory descriptive statistical analysis was done using Python version 3.9 and the formulas described by Ying et al.[21] Visualization of the confusion matrices was done using Python, version 3.9, and the seaborn library (https://www.python.org/). To compare the performance of observed and expected dignity classifications of our multispectral image analysis model under two different illumination modalities, a chi-square test was applied to the confusion matrices. The null hypothesis assumes no significant association between predicted and actual classifications. If rejected, the model's predictions are indicated to have some predictive power. Chi-square test calculations were performed using Graph Pad Prism Version 10.2.3. (https://www.graphpad.com/). P-values < 0.05 were considered statistically significant. Statistical methods were reviewed and validated by a statistician of the Department of Clinical Research at the University Hospital Basel.

Ethics statement
The study was conducted in accordance with the Declaration of Helsinki and approved by the Northwest and Central Switzerland Ethics Committee (2021-02292). The patients in this manuscript have given written informed consent to publication of their case details.

Results
Patient demographics and lesion characteristics
A total of 30 lesions on 24 patients were imaged for subsequent classification. Two lesions were excluded due to lack of histopathological correlation (patient died before excision), and one lesion due to imaging with incorrect exposure times. Therefore, our final data analysis included 27 lesions from 23 patients. Demographic and lesion characteristics are displayed in Table 1.

Lesion classification
For incident light mode analysis, the developed model was applied to 23 lesions (4 excluded due to incorrect imaging), resulting in 6 malignant and 17 benign lesions. An image exposure time of 100 ms was used for standardized comparison. 12 lesions were classified as malignant and 11 as benign, achieving a sensitivity of 83.3% and specificity of 58.8% in comparison to the true dignity determined by histopathological examination (Fig. 5A). Using incident light images as input data, there was no statistically significant association between actual and predicted classification (Chi-square with Yates' correction: $p = 0.193$).

In comparison, direct coupling light mode with spectral analysis of the same 23 lesions resulted in a superior sensitivity of 100% and specificity of 82.4% (classification: 9 malignant, 14 benign) (Fig. 5B). Statistically, the model's predictions were significantly associated with the histological ground truth dignity (Chi-square with Yates' correction: $p = 0.002$). Incorporating the 4 previously excluded lesions, for which data in the direct coupling light mode was available, specificity reduced slightly to 76.2% (sensitivity 100% classification: 11 malignant, 16 benign; $p = 0.004$) (Fig. 5C).

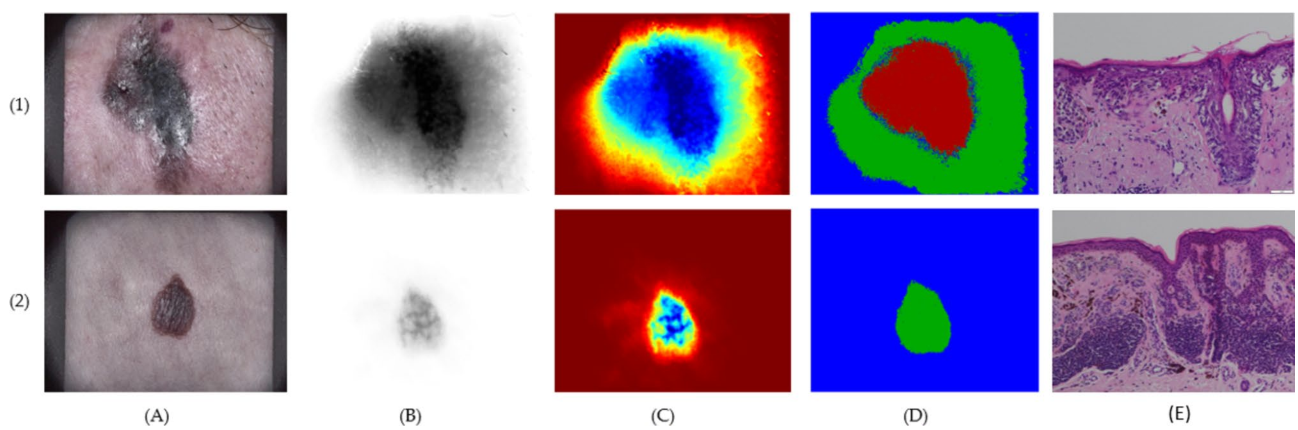

Figure 4. Comparative overview of multispectral image analysis of a melanoma (**1**) and a benign melanocytic nevus (**2**). (**A**) RGB image for lesion localization; (**B**) Snapshot-mosaic raw image; (**C**) Image after pre-processing in fluxTrainer; (**D**) Lesion classification result: Blue—healthy skin, Green—benign melanocytic, Red—malignant melanoma. (**E**) Histology: (**1**) Superficial spreading melanoma, Stage IB (pT2a, Breslow 1.7 mm). Stage classification according to the American Joint Committee on Cancer (AJCC) (8th edition); (**2**) Melanocytic compound-nevus with pigment incontinence.

Number of patients (n)	23
Mean age (years, SD)	56.9 (± 16.4)
Gender (n, %)	
Male	15 (65.2%)
Female	8 (34.8%)
Fitzpatrick skin type	
I	3 (13.0%)
II	9 (39.1%)
III	11 (47.8%)
Previous melanoma (n, %)	
Yes	8 (34.8%)
No	15 (65.2%)
Melanoma risk factor group (n, %)	
Multiple melanocytic naevi (≥ 100) and/or dysplastic nevi (≥ 5) and/or positive family history for melanoma	9 (39.1%)
Personal history of melanoma	8 (34.8%)
No risk factor	6 (26.1%)
Active smoker (n, %)	
Yes	4 (17.4%)
No	19 (82.6%)
spO2 (%, SD)	97.2 (± 1.2)
Concomitant dermatological conditions (n, %)	
Yes	11 (47.8%)
No	12 (52.2%)
Type of concomitant dermatological condition (n, %)[1]	
Non-melanoma skin cancer (including actinic keratosis)	4 (36.4%)
Seborrheic dermatitis	2 (18.2%)
Psoriasis vulgaris	1 (9.1%)
Rosacea	1 (9.1%)
Prurigo simplex subacuta	1 (9.1%)
Onychomycosis	1 (9.1%)
Folliculitis	1 (9.1%)
Suspicion of scabies infestation	1 (9.1%)
Suspicion of acute urticaria	1 (9.1%)
Number of included lesions (n)[2]	27
Anatomical localization (n, %)	
Head/neck	4 (14.8%)
Trunk	18 (66.7%)
Upper extremities	2 (7.4%)
Lower extremities	3 (11.1%)
Histopathological results	
Melanoma	6* (22.2%)
Melanoma subtypes	
Superficial spreading	5 (83.3%)
Borderline Lentigo maligna	1* (16.7%)
AJCC melanoma stage classification (8th edition)	
0	1* (16.7%)
IA	3 (50.0%)
IB	1 (16.7%)
IIA	0 (0.0%)
IIB	1 (16.7%)
Continued	

IIC–IV	0 (0.0%)
Dysplastic melanocytic nevus	6 (22.2%)
Melanocytic nevus	12 (44.4%)
Other[3]	3 (11.1%)

Table 1. Patient demographics and characteristics of included lesions. *AJCC* American Joint Committee on Cancer. [1]Multiple conditions can be stated for a single patient. [2]3 lesions excluded due to patient death before histopathological examination (n = 2) and false imaging exposure times (n = 1). [3]Irritated pigmented seborrheic keratosis (n = 1), blue nevus (n = 1), dermatofibroma (n = 1). *One lesion borderline malignant: Atypical lentiginous melanocytic proliferation in actinically damaged skin for which an incipient transition into an initial lentigo maligna cannot be ruled out with certainty.

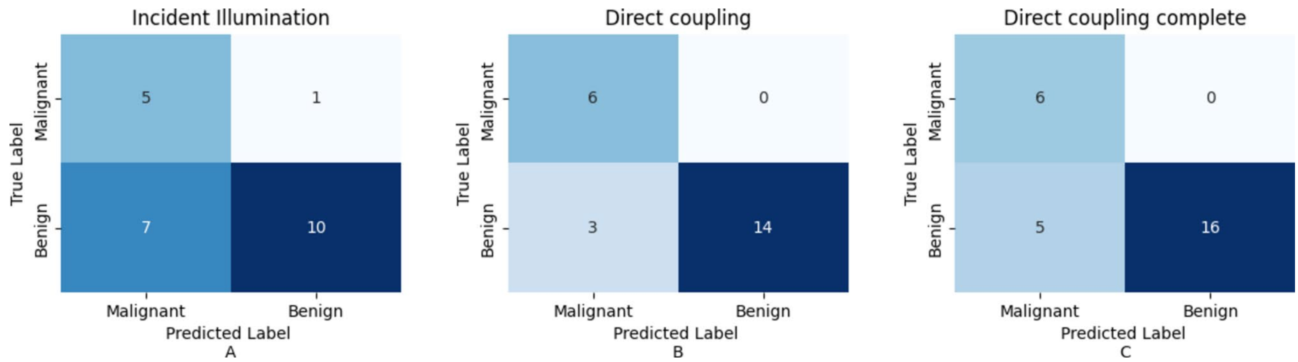

Figure 5. Confusion matrices of (sensitivity, specificity): (**A**) Incident illumination mode, n = 23 (83.3%, 58.8%; p = 0.193); (**B**) direct coupling of light mode, n = 23 (100%, 82.4%; p = 0.002); (**C**) Direct coupling of light, n = 27 (100%, 76.2%; p = 0.004). Statistical analysis of confusion matrices were performed with Chi-square test using Yates' correction.

Our hypothesis that direct illumination can extract more information from the skin was therefore supported by a 16.7% increase in diagnostic sensitivity (incident: 83.3%, direct: 100%) and a 23.6% increase in specificity (incident: 58.8%, direct: 82.4%).

When comparing the accuracy results of multispectral direct light coupling with the classification of RGB lesion images using a commercially-available CNN (MoleAnalyzer-Pro, Version 6.0.5, FotoFinder ATBM® Systems GmbH, Bad Birnbach, Germany, https://www.fotofinder.de/), multispectral imaging outperformed RGB classification. The MoleAnalyzer-Pro analysis showed a 16.7% lower sensitivity (83.3%), detecting 5 out of 6 malignant lesions. The specificity of RGB CNN image analysis was 20.9% lower (61.5%) compared to direct light coupling multispectral image classification.

Histopathological correlation

Areas with maximum histopathological regression corresponded to shiny white areas as seen in RGB images and multispectral images of malignant lesions (Figs. 2B–E, 4). This feature was not observed in benign lesions.

Discussion

To facilitate early detection of melanoma, non-invasive examination methods such as multispectral imaging of melanocytic lesions can assist dermatologists in clinical decision-making. To our knowledge, our study is the first analysis using direct light coupling for multispectral imaging in dermatology. When imaging melanocytic lesions with a spatially and spectrally resolving snapshot mosaic camera prototype, switching the illumination mode from incident light to direct light resulted in superior accuracy of subsequent dignity classification. As parameters such as exposure time were standardised, identical lesions were imaged for comparison and classification models were built in the same way, only the type of illumination during image acquisition varied. Therefore, the observed superior multispectral image classification performance with direct light coupling can be directly attributed to this newly developed illumination modality.

Compared to the 100% accuracy achieved by multispectral direct light coupling image classification, CNN classification of RGB images by MoleAnalyzer-Pro detected only 5 out of 6 malignant lesions. The missed lesion was considered borderline malignant after histopathological examination (atypical lentiginous melanocytic proliferation for which initial lentigo maligna could not be excluded). This lesion was also misclassified using incident light imaging data. Since correct malignant classification was achieved only with direct light coupling, we conclude that the novel illumination mode has the potential to improve early melanoma detection by identifying early features of malignancy. This is supported by the significant statistical predictive power of the model's dignity classification using direct illumination images as input data, while no significance was observed using incident light images. Given the additional superior specificity of direct light coupling multispectral image classification compared to RGB CNN performance, the use of this novel technology could potentially reduce unnecessary excisions for patients.

The accuracies reported in this study were calculated using images with standardised exposure times (100 ms). Shorter exposure times (100 ms or 130 ms) performed better than longer exposure times (150 ms or 300 ms), possibly due to overexposure of the multispectral images.

Evaluation of diagnostic performance

Evaluation of the diagnostic performance of our camera prototype is promising. German dermatologists have been shown to achieve an average sensitivity of 74.1% and specificity of 60% for melanoma classification using dermoscopic images[22]. Our prototype exceeded these diagnostic accuracy rates, suggesting that multispectral imaging can be a useful tool to assist clinicians. Especially primary care physicians and dermatology residents with limited dermoscopic experience could potentially benefit from novel technologies for more accurate skin cancer detection[23].

The benefits and potential applications of multispectral imaging for skin cancer classification are currently under intense investigation[24]. MelaFind, a previously approved device by the United States Food and Drug Administration for melanoma detection using proprietary computer-guided multispectral imaging with 10 spectral wavelengths from 430 to 950 nm, was withdrawn from sale in 2017[25,26]. Key criticisms included a low specificity of 9.5%, resulting in unnecessary biopsies[26,27]. A clinical investigation of Mela Find's diagnostic performance demonstrated a sensitivity of 100%, specificity of 5.5% and a positive predictive value (PPV) of 2.8%[28]. Comparing this performance with preliminary data from our direct light camera prototype shows that the specificity of our device is 15 times higher. However, due to study design heterogeneity (pilot study vs. large prospective trial), no direct conclusion of superiority can be drawn.

Applications of our multispectral device may extend beyond melanoma detection to recognition of other skin cancers. Delpueyo et al. used multispectral imaging to detect melanoma and basal cell carcinomas in 429 lesions[29]. While the authors reported a sensitivity of 91.3% and a specificity of 54.5% for basal cell carcinoma, we have not tested our device on non-melanoma skin cancers, but believe that successful malignant classification would be possible based on the underlying mechanism of action. In the classification of melanocytic lesions, the sensitivity (87.2%) and specificity (54.5%) for melanoma were again exceeded by our device, but comparison is limited by significant cross-trial differences[29].

The use of CNNs to classify pigmented skin lesions demonstrates promising performance[30]. A pilot study by Lindholm et al. on 3D hyperspectral imaging combined with CNN assessment of 20 pigmented skin lesions showed a sensitivity and specificity of 95% and 97% using a majority voting method[31]. In comparison, our model achieves a slightly higher sensitivity but significantly lower specificity. We attribute this to the classification method used, as the authors also included few lesions, but compensated with their deep learning method of processing and classification. We suggest that a new generation of multispectral imaging and classification models could be achieved by combining our novel direct illumination mode with subsequent CNN analysis. This would provide a new tool for augmented intelligence, the combination of artificial and human intelligence, which has shown great diagnostic potential in a previous clinical study investigating CNN classification of digital dermoscopic images in real-world application[30].

In addition to improved classification, both multispectral and RGB images in our study showed potential to aid clinician decision-making by macroscopically highlighting underlying histopathological features such as regression in the form of white streaks, which were only observed in malignant lesions. This presence of shiny white streaks has previously been identified as a sign of malignancy with an odds ratio of 10.5 in polarized dermoscopy of pigmented skin lesions, leading clinicians to biopsy to exclude melanoma[32].

Strengths and limitations

To our knowledge, our study is the first to demonstrate that direct illumination improves the diagnostic accuracy of multispectral imaging, providing a basis for further improvement in dermatological applications. A major strength of this study is that the prototype is a snapshot mosaic camera, which allows fast image acquisition without delay between different wavelengths. In addition, the prototype is handheld, which allows for fast acquisition and reduces the risk of potential noise from application or patient movement. Because the prototype is portable and lightweight, it could be widely used in routine clinical practice after further validation. As multispectral imaging is also being investigated for the detection of other solid tumour types[33], the direct light coupling method could potentially improve cancer detection and classification in other medical fields.

Since our pilot study was designed to evaluate a novel illumination method as a proof of concept and not to validate a camera prototype, several major limitations must be considered. First, the small number of included lesions severely limits the generalisability and comparability of the results. Larger, statistically powered validation studies with more lesions need to be performed before definitive conclusions about the superiority of illumination methods can be drawn. Unfortunately, four images were acquired incorrectly in the incident mode of illumination, leaving only 23 lesions for illumination modality comparison. When the sensitivity and specificity of direct light coupling were recalculated using the available data from 27 lesions, the sensitivity remained unchanged while the specificity decreased by 6.2%. With more data to train and test the model, the results obtained could potentially become more robust. Due to this aspect and general differences between study designs, comparisons with other devices are severely limited. By selecting lesions already suspected of being malignant, our study additionally risks verification bias, potentially skewing specificity estimates. In addition, the lesion image classification model was not built using open-source software such as Python. Using a CNN for classification could potentially further improve the results obtained. In future research, it is essential to apply our direct light multispectral imaging approach to a larger, more diverse sample including clearly benign lesions, with verification by sequential dermoscopic monitoring of benign lesions to assess the technology's specificity more accurately. Additionally, conducting head-to-head comparisons with existing diagnostic tools and evaluations by dermatologists are necessary to gain a clearer understanding of any potential advancements.

Conclusion

In summary, we have developed a spatially and spectrally resolving snapshot mosaic camera prototype using a novel method of direct illumination to image and classify melanocytic lesions. As a proof-of-concept, we have demonstrated in a limited sample size that direct coupling of light allows more information to be extracted from deeper skin layers for a more accurate lesion classification compared to incident illumination. In addition, we found that the use of multispectral images for lesion classification outperformed CNN RGB image classification. Coupling direct light multispectral imaging with a CNN could potentially provide a powerful non-invasive diagnostic tool for cancer detection in dermatology and other medical fields. Further studies with more lesions are needed to validate the camera prototype.

Data availability

The data underlying this article will be shared on reasonable request to the corresponding authors.

Received: 26 May 2024; Accepted: 8 August 2024
Published online: 16 August 2024

References

1. Garbe, C. *et al.* Epidemiology of cutaneous melanoma and keratinocyte cancer in white populations 1943–2036. *Eur. J. Cancer* **152**, 18–25. https://doi.org/10.1016/j.ejca.2021.04.029 (2021).
2. Schadendorf, D. *et al.* Melanoma. *Lancet* **392**(10151), 971–984. https://doi.org/10.1016/S0140-6736(18)31559-9 (2018).
3. Rey-Barroso, L. *et al.* Optical technologies for the improvement of skin cancer diagnosis: A review. *Sensors* **21**(1), 252. https://doi.org/10.3390/s21010252 (2021).
4. Aggarwal, L. P. & Papay, F. A. Applications of multispectral and hyperspectral imaging in dermatology. *Exp. Dermatol.* **31**(8), 1128–1135. https://doi.org/10.1111/exd.14624 (2022).
5. Johansen, T. H. *et al.* Recent advances in hyperspectral imaging for melanoma detection. *WIREs Comput. Stat.* **12**(1), 1465. https://doi.org/10.1002/wics.1465 (2020).
6. Leon, R. *et al.* Non-invasive skin cancer diagnosis using hyperspectral imaging for in-situ clinical support. *J. Clin. Med.* **9**(6), 662. https://doi.org/10.3390/jcm9061662 (2020).
7. Ilișanu, M. A., Moldoveanu, F. & Moldoveanu, A. Multispectral imaging for skin diseases assessment—State of the art and perspectives. *Sensors* **23**(8), 888. https://doi.org/10.3390/s23083888 (2023).
8. Halicek, M., Fabelo, H., Ortega, S., Callico, G. M. & Fei, B. In-vivo and ex-vivo tissue analysis through hyperspectral imaging techniques: Revealing the invisible features of cancer. *Cancers* **11**(6), 756. https://doi.org/10.3390/cancers11060756 (2019).
9. Bekina, A. *et al.* Multispectral assessment of skin malformations using a modified video-microscope. *Latv. J. Phys. Tech. Sci.* **49**(5), 4–8. https://doi.org/10.2478/v10047-012-0024-2 (2012).
10. Nishidate, I., Maeda, T., Niizeki, K. & Aizu, Y. Estimation of melanin and hemoglobin using spectral reflectance images reconstructed from a digital RGB image by the Wiener estimation method. *Sensors* **13**(6), 7902–7915. https://doi.org/10.3390/s130607902 (2013).
11. Aloupogianni, E. *et al.* Design of a hyper-spectral imaging system for gross pathology of pigmented skin lesions. In *2021 43rd Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC)* 3605–3608. https://doi.org/10.1109/EMBC46164.2021.9629512 (IEEE, 2021).
12. Rey-Barroso, L. *et al.* Visible and extended near-infrared multispectral imaging for skin cancer diagnosis. *Sensors* **18**(5), 1441. https://doi.org/10.3390/s18051441 (2018).
13. Zherebtsov, E. *et al.* Hyperspectral imaging of human skin aided by artificial neural networks. *Biomed. Opt. Express* **10**(7), 3545. https://doi.org/10.1364/BOE.10.003545 (2019).
14. Meffert, B. & Meffert, H. Optische Strahlung und ihre Wirkungen auf die Haut—Optical radiation and its effects on the skin. *Biomed. Tech. Eng.* **45**(4), 98–104. https://doi.org/10.1515/bmte.2000.45.4.98 (2000).
15. Fuchs, T. & Weber, S. *Vorrichtung zur Abbildungserzeugung von Hautläsionen*. https://patents.google.com/patent/DE102019113283A1 (2019).
16. Dittrich, P.-G., Bichra, M., Pfützenreuter, C., Rosenberger, M. & Notni, G. Measurement principle and arrangement for the determination of spectral channel-specific angle dependencies for multispectral resolving filter-on-chip CMOS cameras. *SPIE Int. Soc. Opt. Eng.* **2**, 871. https://doi.org/10.1117/12.2527871 (2019).
17. Dittrich, P. G., Kraus, D., Ehrhardt, E., Henkel, T. & Notni, G. Multispectral imaging flow cytometry with spatially and spectrally resolving snapshot-mosaic cameras for the characterization and classification of bioparticles. *Micromachines* **13**(2), 238. https://doi.org/10.3390/mi13020238 (2022).
18. Dittrich, P.-G., Rosenberger, M. & Notni, G. Measurement, comparison, and evaluation of spectral sensitivity curves from different multispectral resolving snapshot-mosaic cameras (Conference Presentation). In *Algorithms, Technologies, and Applications for Multispectral and Hyperspectral Imagery XXVI* (eds. Messinger, D. W. & Velez-Reyes, M.) 63. https://doi.org/10.1117/12.2564690 (SPIE, 2020).
19. Akima, H. A new method of interpolation and smooth curve fitting based on local procedures. *J. ACM* **17**, 589–660 (1970).
20. Ruffin, C. & King, R. L. *The Analysis of Hyperspectral Data Using Savitzky–Golay Filtering-Theoretical Basis. 1* 756–758 (Institute of Electrical and Electronics Engineers, 2003).
21. Ying, G.-S., Maguire, M. G., Glynn, R. J. & Rosner, B. Calculating sensitivity, specificity, and predictive values for correlated eye data. *Investig. Opthalmol. Vis. Sci.* **61**(11), 29. https://doi.org/10.1167/iovs.61.11.29 (2020).
22. Brinker, T. J. *et al.* Deep learning outperformed 136 of 157 dermatologists in a head-to-head dermoscopic melanoma image classification task. *Eur. J. Cancer* **113**, 47–54. https://doi.org/10.1016/j.ejca.2019.04.001 (2019).
23. Goessinger, E. V. *et al.* Patient and dermatologists' perspectives on augmented intelligence for melanoma screening: A prospective study. *J. Eur. Acad. Dermatol. Venereol.* **1**, 1–10. https://doi.org/10.1111/jdv.19905 (2024).
24. Aloupogianni, E., Ishikawa, M., Kobayashi, N. & Obi, T. Hyperspectral and multispectral image processing for gross-level tumor detection in skin lesions: A systematic review. *J. Biomed. Opt.* **27**(06), 901. https://doi.org/10.1117/1.jbo.27.6.060901 (2022).
25. Beltrami, E. J. *et al.* Artificial intelligence in the detection of skin cancer. *J. Am. Acad. Dermatol.* **87**(6), 1336–1342. https://doi.org/10.1016/j.jaad.2022.08.028 (2022).

26. Fink, C. & Haenssle, H. A. Non-invasive tools for the diagnosis of cutaneous melanoma. *Skin Res. Technol.* **23**(3), 261–271. https://doi.org/10.1111/srt.12350 (2017).
27. Monheit, G. *et al.* The performance of MelaFind: A prospective multicenter study. *Arch. Dermatol.* **147**(2), 188–194. https://doi.org/10.1001/archdermatol.2010.302 (2011).
28. Fink, C., Jaeger, C., Jaeger, K. & Haenssle, H. A. Diagnoseleistung des MelaFind-Geräts im klinischen Alltag. *J. Ger. Soc. Dermatol.* **15**(4), 414–419. https://doi.org/10.1111/ddg.13220 (2017).
29. Delpueyo, X. *et al.* Multispectral imaging system based on light-emitting diodes for the detection of melanomas and basal cell carcinomas: A pilot study. *J. Biomed. Opt.* **22**(6), 065006. https://doi.org/10.1117/1.jbo.22.6.065006 (2017).
30. Cerminara, S. E. *et al.* Diagnostic performance of augmented intelligence with 2D and 3D total body photography and convolutional neural networks in a high-risk population for melanoma under real-world conditions: A new era of skin cancer screening? *Eur. J. Cancer* **190**, 954. https://doi.org/10.1016/j.ejca.2023.112954 (2023).
31. Lindholm, V. *et al.* Differentiating malignant from benign pigmented or non-pigmented skin tumours—A pilot study on 3D hyperspectral imaging of complex skin surfaces and convolutional neural networks. *J. Clin. Med.* **11**(7), 1914. https://doi.org/10.3390/jcm11071914 (2022).
32. Shitara, D. *et al.* Shiny white streaks: A sign of malignancy at dermoscopy of pigmented skin lesions. *Acta Derm. Venereol.* **94**(2), 132–137. https://doi.org/10.2340/00015555-1683 (2014).
33. Zhang, Y. *et al.* Applications of hyperspectral imaging in the detection and diagnosis of solid tumors. *Transl. Cancer Res.* **9**(2), 1265–1277. https://doi.org/10.21037/tcr.2019.12.53 (2020).

Acknowledgements
The authors thank the patients who participated in this study. The authors gratefully acknowledge the financial support from the German Federal Ministry of Education and Research. Furthermore, we acknowledge support for the publication costs by the Open Access Publication Fund of the Technische Universität Ilmenau.

Author contributions
Conceptualization, P.-G.D. and L.V.M.; methodology, P.-G.D.; software, P.-G.D. and S.W.; validation, P.-G.D. and G.N.; formal analysis, P.-G.D. and P.N.; investigation, E.G., B.M. and L.V.M.; resources, P.-G.D. and L.V.M..; data curation, P.-G.D.; writing—original draft preparation, P.N., P.-G.D., E.G. and L.V.M.; writing—review and editing, P.N., P.-G.D., E.G., S.C., B.M., A.A.N. and L.V.-M.; visualization, P.-G.D.; supervision, L.V.M., and G.N.; project administration, P.-G.D. and L. V.M.; funding acquisition, P.-G.D. All authors have read and agreed to the published version of the manuscript.

Funding
This work was funded by the Federal Ministry of Education and Research through the support program for digital optics with its project "MINIMIZE", Grant No. 13N14835 in the framework of Photonic Research Germany.

Competing interests
The authors declare no conflicts of interest in the subject or materials discussed in this manuscript. Unrelated to the present manuscript, AAN declares being a consultant and advisor and/or receiving speaking fees and/or grants and/or served as an investigator in clinical trials for AbbVie, Almirall, Amgen, Biomed, BMS, Boehringer Ingelheim, Celgene, Eli Lilly, Galderma, GSK, LEO Pharma, Janssen-Cilag, MSD, Novartis, Pfizer, Pierre Fabre Pharma, Regeneron, Sandoz, Sanofi, and UCB.

Additional information

Correspondence and requests for materials should be addressed to P.-G.D. or L.V.M.